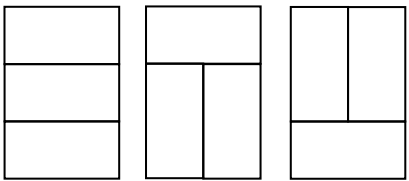
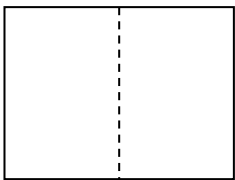
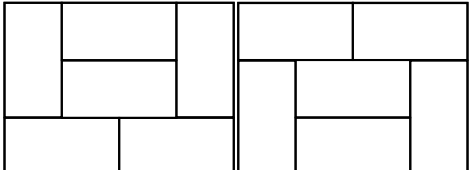
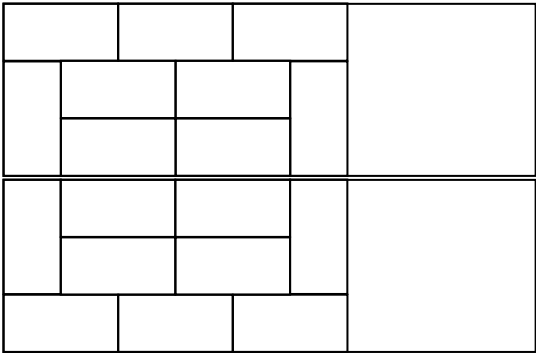


2022 H3 Math Prelim Marking Scheme

Qn	Suggested Solution	
1	Let $k = ad - bc$. Suppose a, b, c and d are all divisible by k. Then, $k^2 \mid ad \text{ and } k^2 \mid bc$ $\Rightarrow k^2 \mid ad - bc$ $\Rightarrow k^2 \mid k$ $\Rightarrow k = \pm 1$ $\Rightarrow ad - bc = \pm 1,$ contradicting $ad - bc > 1$. $\square$	

2(i)	Draw 3 cases for u_1.  For u_2: Case 1: Divide 3×4 rectangle into two 3×2 rectangles, there are $u_1^2 = 9$ ways.  Case 2: There is at least one rectangle placed across the vertical dividing line in Case 1. Proof by exhaustion gives two more ways. (One way is to divide this case according to the orientation of the rectangle in the top left corner.) 	
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(ii)	We start from one side of the rectangle and consider where is the nearest vertical dividing line parallel to that side. Case 1: The rectangle is divided into a 3×2 rectangle and a $3 \times (2(r-1))$ rectangle. There are $3u_{r-1}$ ways.  For all integers k such that $2 \leq k \leq r-1$, we consider Case k: The rectangle is divided into a $3 \times (2k)$ rectangle (which does not contain any vertical dividing lines) and a $3 \times (2(r-k))$ rectangle. Note that there are only two ways to cover the $3 \times (2k)$ rectangle that does not contain any vertical dividing lines. (One way is to prove this using the orientation of the rectangle in the top left corner.) There are u_{r-k} ways to cover the rest of the rectangle. Hence, there are $2u_{r-k}$ ways. (Diagram shows examples for $k = 3$.)  Case r: The rectangle does not have any dividing lines. There are 2 ways.	
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	Hence, the total number of ways is $3u_{r-1} + \sum_{k=2}^{r-1} 2u_{r-k} + 2$ $= 3u_{r-1} + 2 + \sum_{k=1}^{r-2} 2u_k$	
(iii)	Eq (1): $u_r = 3u_{r-1} + 2 + 2\sum_{k=1}^{r-2} u_k$ Eq (2): $u_{r-1} = 3u_{r-2} + 2 + 2\sum_{k=1}^{r-3} u_k$ (1) – (2): $u_r - u_{r-1} = 3u_{r-1} - 3u_{r-2} + 2\sum_{k=1}^{r-2} u_k - 2\sum_{k=1}^{r-3} u_k$ $u_r = 4u_{r-1} - 3u_{r-2} + 2u_{r-2}$ $u_r = 4u_{r-1} - u_{r-2}$	

3(i)	Suppose the boxes are distinguishable. We solve this by complementation principle. To put n objects in 2 distinguishable boxes without restrictions, there are 2^n ways. To put n objects in 2 distinguishable boxes with one empty box, there are 2 ways. To put n objects in 2 distinguishable boxes with one box containing exactly one object, there are $2n$ ways. Hence there are $2^n - (2n + 2)$ ways to put n objects in 2 distinguishable boxes such that there are at least 2 objects in each box. If the boxes are indistinguishable, we divide by $2!$ to get $S_2(n, 2) = 2^{n-1} - n - 1$	
(ii)	Suppose the boxes are distinguishable. We solve this by complementation principle. To put n objects in 3 distinguishable boxes without restrictions, there are 3^n ways. There are 3 cases in the complement (at least one box contains at most one object): <u>Case 1: No empty boxes</u> To put n objects in 3 distinguishable boxes such that two boxes contain exactly one object, there are $3n(n-1)$ ways. To put n objects in 3 distinguishable boxes such that only one box contains exactly one object (and the other two boxes both contain at least one object), we select one object and one box ($3n$ ways) and put the remaining $n-1$ objects in the remaining 2 boxes such that no box is empty ($2^{n-1} - 2$ ways). Hence, there are $3n(2^{n-1} - 2)$ ways. But each case where two boxes contain exactly one object is double counted here. Hence, there are $3n(2^{n-1} - 2) - 3n(n-1)$ $= 3n(2^{n-1} - n - 1)$ ways in this case.	

	<u>Case 2: Exactly one empty box</u> If one of the boxes is empty (3 ways), then we want to put the objects into 2 boxes such that there is at least one item in each box ($2^n - 2$ ways). Hence, there are $3(2^n - 2)$ ways. <u>Case 3: Two empty boxes</u> There are 3 ways. Hence there are $3^n - 3n(2^{n-1} - n - 1) - 3(2^n - 2) - 3$ $= 3^n - 3n(2^{n-1} - n - 1) - 3(2^n) + 3$ ways to put n objects in 3 $= 3^n - 3(2^{n-1})(n+2) + 3(n^2 + n + 1)$ distinguishable boxes such that there are at least 2 objects in each box. If the boxes are indistinguishable, we divide by 3! to get $S_3(n, 3) = \frac{3^{n-1} + n^2 + n + 1}{2} - (n+2)2^{n-2}$	
(iii)	If n is even, $S_2(n, \frac{1}{2}n)$ is the number of ways to put exactly 2 objects in each box. $S_2(n, \frac{1}{2}n) = \frac{\binom{n}{2} \binom{n-2}{2} \binom{n-4}{2} \dots \binom{4}{2} \binom{2}{2}}{(\frac{1}{2}n)!}$ $= \frac{\frac{n(n-1)}{2} \frac{(n-2)(n-3)}{2} \dots \frac{(4)(3)}{2} \frac{(2)(1)}{2}}{(\frac{1}{2}n)!}$ $= \frac{[(n-1)(n-3) \dots (3)(1)] \left[\frac{n(n-2)}{2} \frac{(4)(2)}{2} \right]}{(\frac{1}{2}n)!}$ $= (n-1)(n-3) \dots (3)(1) \frac{(\frac{1}{2}n)(\frac{1}{2}n-1) \dots (2)(1)}{(\frac{1}{2}n)!}$ $= (n-1)(n-3) \dots (3)(1) \frac{(\frac{1}{2}n)!}{(\frac{1}{2}n)!}$ $= (n-1)(n-3) \dots (3)(1)$	
(iv)	If n is even, $S_3(n, \frac{1}{2}(n-1))$ is the number of ways to put exactly 3 objects in one box and 2 objects in each of the other boxes.	

	$S_2\left(n,\frac{1}{2}(n-1)\right)$ $=\binom{n}{3}S_2\left(n-3,\frac{1}{2}(n-3)\right)$ $=\frac{n(n-1)(n-2)}{6}(n-4)(n-6)\cdots(3)(1)$	
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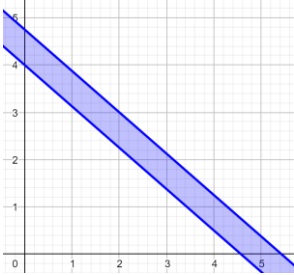
4(a)	$\int \frac{x}{(x^2-4)\sqrt{1-x^2}} dx = \int \frac{\sin u}{\sin^2 u - 4} du$ $= \int \frac{-\sin u}{3 + \cos^2 u} du$ $= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\cos u}{\sqrt{3}} \right) + c$ $= \frac{1}{\sqrt{3}} \tan^{-1} \left(\sqrt{\frac{1-x^2}{3}} \right) + c$	
(b) (i)	$I_1 + I_2 = \int_0^k 1 dx = k$ $I_2 - I_1 = \frac{1}{k} \int_0^k \frac{k \cos kx - k \sin kx}{4 + \sin kx + \cos kx} dx$ $= \frac{1}{k} \left[\ln(4 + \sin kx + \cos kx) \right]_0^k$ $= \frac{1}{k} \left[\ln(4 + \sin(k^2) + \cos(k^2)) - \ln 5 \right]$ Therefore, $I_1 = \frac{1}{2} \left[(I_1 + I_2) - (I_2 - I_1) \right]$ $= \frac{1}{2k} \left[k^2 - \ln \frac{4 + \sin(k^2) + \cos(k^2)}{5} \right]$	
(ii)	Let $I_3 = \int_0^\pi \frac{3 + \cos x}{13 + 2 \sin x + 3 \cos x} dx$ $I_4 = \int_0^\pi \frac{2 + \sin x}{13 + 2 \sin x + 3 \cos x} dx.$ $3I_3 + 2I_4 = \int_0^\pi \frac{9 + 3 \cos x + 4 + 2 \sin x}{13 + 2 \sin x + 3 \cos x} dx = \pi$ $2I_3 - 3I_4 = \int_0^\pi \frac{6 + 2 \cos x - 6 - 3 \sin x}{13 + 2 \sin x + 3 \cos x} dx$ $= \left[\ln(13 + 2 \sin x + 3 \cos x) \right]_0^\pi$ $= \ln 10 - \ln 16 = \ln \frac{5}{8}$ So $I_3 = \frac{3(3I_3 + 2I_4) + 2(2I_3 - 3I_4)}{13} = \frac{1}{13} \left(3\pi + 2 \ln \frac{5}{8} \right).$	

5 (a)	$\frac{d}{dx}(1+x+x^2+\dots+x^n) = \frac{d}{dx}\left(\frac{1-x^{n+1}}{1-x}\right)$ $1+2x+3x^2+\dots+nx^{n-1} = \frac{1-(n+1)x^n+nx^{n+1}}{(1-x)^2}$ Sub $x = -1$, $1-2+3-4+5-6+\dots+(-1)^{n-1}n = \frac{1}{4}\left[1-(n+1)(-1)^n+n(-1)^{n+1}\right]$ $= \begin{cases} -\frac{n}{2}, & \text{when } n \text{ is even} \\ \frac{n+1}{2}, & \text{when } n \text{ is odd} \end{cases}$	
(b)	When n is even, $(1^2-2^2)+(3^2-4^2)+(5^2-6^2)+\dots+(n^2-n^2)$ $= (1-2)(1+2)+(3-4)(3+4)+(5-6)(5+6)+\dots+(-1)(n-1+n)$ $= -[1+2+3+4+5+6+\dots+n]$ $= -\frac{n(n+1)}{2}$ When n is odd, $(1^2-2^2)+(3^2-4^2)+\dots+(n-2)^2-(n-1)^2+n^2$ $= -[1+2+3+4+5+6+\dots+(n-1)]+n^2$ $= -\frac{(n-1)(n)}{2}+n^2$ $= \frac{n(n+1)}{2}$ For both cases, the sum can be expressed as $\frac{(-1)^{n+1} \cdot n(n+1)}{2}$. <u>Alternatively</u> $1+2x+3x^2+\dots+nx^{n-1} = \frac{1-(n+1)x^n+nx^{n+1}}{(1-x)^2}$ $\Rightarrow x(1+2x+3x^2+\dots+nx^{n-1}) = \frac{x(1-(n+1)x^n+nx^{n+1})}{(1-x)^2}$ Differentiating with respect to x, $1^2+2^2x+3^2x^2+\dots+n^2x^{n-1}$ $= \frac{(1-x)^2[1-(n+1)^2x^n+n(n+2)x^{n+1}]+2(1-x)x[1-(n+1)x^n+nx^{n+1}]}{(1-x)^4}$	

	When $x = -1$, $1^2 - 2^2 + \dots + (-1)^{n-1} n^2$ $= \frac{4[1 + (-1)^{n+1}(n+1)^2 + n(n+2)(-1)^{n+1}] - 4[1 + (-1)^{n+1}(n+1) + (-1)^{n+1}n]}{16}$ $= \frac{(-1)^{n+1}[(n+1)^2 + n^2 + 2n - (n+1) - n]}{4}$ $= \frac{(-1)^{n+1}(2n^2 + 2n)}{4}$ $= \frac{1}{2}(-1)^{n+1}n(n+1)$	
(c)	When n is odd, $1^3 - 2^3 + 3^3 - 4^3 + 5^3 - 6^3 + \dots + n^3$ $= [1^3 + 2^3 + \dots + n^3] - 2[2^3 + 4^3 + 6^3 + \dots + (n-1)^3]$ $= \left(\frac{n(n+1)}{2}\right)^2 - 2\left(2^3\right)\left[1^3 + 2^3 + 3^3 + \dots + \left(\frac{n-1}{2}\right)^3\right]$ $= \left(\frac{n(n+1)}{2}\right)^2 - 16\left(\frac{\left(\frac{n-1}{2}\right)\left(\frac{n+1}{2}\right)}{2}\right)^2$ $= \frac{1}{4}(n+1)^2[n^2 - (n-1)^2]$ $= \frac{1}{4}(n+1)^2(2n-1)$	

6(i)	Let P_n be the statement $p_n(x+1) - p_n(x) = nx^{n-1}$ for positive integers n. Prove that base case $n = 1$ is true: $\frac{dp_1(x)}{dx} = np_0(x) \Rightarrow p_1(x) = \int 1 dx = x + c_1.$ Therefore, $\text{LHS} = p_1(x+1) - p_1(x) = (x+1+c_1) - (x+c_1) = 1 = 1x^{1-1} = \text{RHS}.$ P_1 is true. Assume that P_k is true for some positive integer k, i.e. $p_k(x+1) - p_k(x) = kx^{k-1}.$ We want to prove that P_{k+1} is true. $\frac{dp_{k+1}(x)}{dx} = (k+1)p_k(x) \Rightarrow p_{k+1}(x) = (k+1) \int p_k(x) dx \quad \text{--- (1)}$ Then $p_{k+1}(x+1) - p_{k+1}(x) = (k+1) \int [p_k(x+1) - p_k(x)] dx$ $= (k+1) \int kx^{k-1} dx \quad (\text{using induction hypothesis})$ $= (k+1)x^k + c_2 \quad \text{--- (2)}$ Since $\int_0^1 p_k(x) dx = 0$, from (1), $\frac{1}{k+1} [p_{k+1}(x)]_0^1 = 0$ $p_{k+1}(1) - p_{k+1}(0) = 0$ Sub $x = 0$ into (2): $p_{k+1}(1) - p_{k+1}(0) = c_2$. Hence $c_2 = 0$. P_k is true $\Rightarrow P_{k+1}$ is true. Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true, by induction P_n is true for all positive integers n.	
(ii)	$p_5(x) = x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{1}{6}x$ $p_6(x) = 6 \int p_5(x) dx$ $= 6 \int \left[x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{1}{6}x \right] dx$ $= x^6 - 3x^5 + \frac{5}{2}x^4 - \frac{1}{2}x^2 + c_3$	

	$\int_0^1 p_6(x) dx = 0$ $\Rightarrow \left[\frac{1}{7}x^7 - \frac{1}{2}x^6 + \frac{1}{2}x^5 - \frac{1}{6}x^3 + c_3x \right]_0^1 = 0$ $\Rightarrow c = \frac{1}{42}$ $\therefore p_6(x) = x^6 - 3x^5 + \frac{5}{2}x^4 - \frac{1}{2}x^2 + \frac{1}{42}$	
(iii)	$p_6(r+1) - p_6(r) = 6r^5 \quad (\text{replace } x \text{ with } r \text{ and let } n = 6)$ $\sum_{r=0}^{N-1} r^5 = \frac{1}{6} \sum_{r=0}^{N-1} [p_6(r+1) - p_6(r)]$ $= \frac{1}{6} [p_6(N) - p_6(N-1)$ $+ p_6(N-1) - p_6(N-2)$ $\vdots$ $+ p_6(2) - p_6(1)]$ $+ p_6(1) - p_6(0)]$ $= \frac{1}{6} [p_6(N) - p_6(0)]$ $= \frac{1}{6} \left(N^6 - 3N^5 - \frac{1}{2}N^4 - \frac{1}{2}N^2 \right)$ Therefore, $\sum_{r=1}^N r^5 = \left(\sum_{r=0}^{N-1} r^5 \right) - 0^5 + N^5 = \frac{1}{6}N^6 + \frac{1}{2}N^5 + \frac{5}{12}N^4 - \frac{1}{12}N^2$ (shown).	

7(i)	(i) $(1+3+5+\dots+(2x-1))+(2+4+6+\dots+2y)\leq 654321$ $\frac{x}{2}(2x)+\frac{y}{2}(2y+2)\leq 654321$ $x^2+(y+0.5)^2\leq 654321.25$	
(ii)_	(ii) $n=21x+24(y+0.5)-12$ $\leq \sqrt{(21^2+24^2)[x^2+(y+0.5)^2]}-12$ $\leq \sqrt{1017(654321.25)}-12$ < 25784.22 Since $n=3(7x+8y)$ is a multiple of 3, we must have $n\leq 25782$.	
(iii)	(iii) x must be odd so that the sum of elements is odd. The only cases are: $x=3, y=2$ and $x=5, y=0$. $x=3, y=2$: possible $S=\{1,2,3,4,654311\}$ $x=5, y=0$: possible $S=\{1,3,5,7,654305\}$ 	

Qn	Suggested Solution	
8 (i)	Let $a = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$ and $b = q_1^{s_1} q_2^{s_2} \dots q_l^{s_l}$ be the (unique) prime factorisation of a and b respectively. So, $\{p_1, p_2, \dots, p_k\} \cap \{q_1, q_2, \dots, q_l\} = \emptyset. \quad \text{----(1)}$ Now, $a^m = p_1^{mr_1} p_2^{mr_2} \dots p_k^{mr_k}$ and $b^n = q_1^{ns_1} q_2^{ns_2} \dots q_l^{ns_l}$. By the fundamental theorem of arithmetic, these are the (unique) prime factorisations of a^m and b^n respectively. From (1), a^m and b^n do not have any common prime factors. Hence $\gcd(a^m, b^n) = 1$. $\square$	
(ii)	Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be a polynomial with integer coefficients and $\frac{p}{q}$ be a rational root of $f(x)$ with $\gcd(p, q) = 1$. Then, $f\left(\frac{p}{q}\right) = 0$. Thus, we have $\left(\frac{p}{q}\right)^n + a_{n-1}\left(\frac{p}{q}\right)^{n-1} + \dots + a_1\left(\frac{p}{q}\right) + a_0 = 0$ $p^n + a_{n-1}p^{n-1}q + \dots + a_1pq^{n-1} + a_0q^n = 0.$ Since $q \mid a_i p^i q^{n-i}$ for all $i = 0, 1, 2, \dots, n-1$, $q \mid p^n$. However, since $\gcd(p, q) = 1$, by (i), $\gcd(p^n, q) = 1$. Together with $q \mid p^n$, it must be the case that $q = \pm 1$, implying that $\frac{p}{q}$ is an integer. $\square$	
(iii)	Let p^k be the smallest power of p that is greater than n, $p^k > n \Rightarrow 0 < \frac{n}{p^k} < 1 \Rightarrow \left\lfloor \frac{n}{p^k} \right\rfloor = 0. \text{ Thus, } \left\lfloor \frac{n}{p^i} \right\rfloor = 0 \text{ for all } i \geq k. \text{ Hence,}$ $\sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor = \sum_{i=1}^{k-1} \left\lfloor \frac{n}{p^i} \right\rfloor + \sum_{i=k}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor = \sum_{i=1}^{k-1} \left\lfloor \frac{n}{p^i} \right\rfloor + 0 = \sum_{i=1}^{k-1} \left\lfloor \frac{n}{p^i} \right\rfloor$ which is a finite series and hence is finite.	

	Now, there are $\left\lfloor \frac{n}{p} \right\rfloor$ multiples of p in $\{1, 2, 3, \dots, n\}$, each of which contribute 1 to the exponent of p in $n!$. - -- (1) Also, there are $\left\lfloor \frac{n}{p^2} \right\rfloor$ multiples of p^2 in $\{1, 2, 3, \dots, n\}$, each of which contribute 1 more to the exponent of p in $n!$ in addition to those already counted in (1). By continuing this process, we see that $e_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \dots + \left\lfloor \frac{n}{p^{k-1}} \right\rfloor$ $= \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \dots + \left\lfloor \frac{n}{p^{k-1}} \right\rfloor + \left\lfloor \frac{n}{p^k} \right\rfloor + \left\lfloor \frac{n}{p^{k+1}} \right\rfloor + \dots$ $\left(\because \left\lfloor \frac{n}{p^i} \right\rfloor = 0 \text{ for all } i \geq k \right)$ $= \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor.$ $e_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor < \sum_{i=1}^{\infty} \frac{n}{p^i} = \frac{n}{p-1} \leq n.$	
(iv)	Suppose the equation $\frac{x^n}{n!} + \frac{x^{n-1}}{(n-1)!} + \frac{x^{n-2}}{(n-2)!} + \dots + \frac{x}{1!} + 1 = 0$ has a rational root, α. Then, $\frac{\alpha^n}{n!} + \frac{\alpha^{n-1}}{(n-1)!} + \dots + \frac{\alpha^m}{m!} + \dots + \frac{\alpha}{1!} + 1 = 0$ $\alpha^n + \frac{n!}{(n-1)!} \alpha^{n-1} + \dots + \frac{n!}{m!} \alpha^m + \dots + \frac{n!}{1!} \alpha + n! = 0 \quad (*)$ By (ii), α is an integer.	

Since $n > 1$, n has a prime divisor, p. Also, $n \mid \frac{n!}{m!}$ for each $m = 0, 1, 2, \dots, n-1$. Hence, $p \mid \frac{n!}{(n-1)!} \alpha^{n-1} + \dots + \frac{n!}{m!} \alpha^m + \dots + \frac{n!}{1!} \alpha + n!.$ From (*), $p \mid \alpha^n$. Since p is a prime, $p \mid \alpha$. Next, for each $m = 0, 1, 2, \dots, n-1$, observe that • $p^m \mid \alpha^m \Rightarrow e_p(\alpha^m) \geq m$ --- (1) and • $e_p(m!) < m$ (by part (iii)). --- (2) Denote $e_p(n!) = r$. --- (3) Then, for each $m = 0, 1, 2, \dots, n-1$, from (1), (2) and (3), we have $\begin{aligned} e_p\left(\frac{n!}{m!} \alpha^m\right) &= e_p(n!) + e_p(\alpha^m) - e_p(m!) \\ &\geq r + m - (m-1) \\ &= r + 1. \end{aligned}$ Together with (*), we have $p^{r+1} \mid n!$ which is a contradiction to (3). $\square$	
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Qn	Suggested Solution	
9(i)	Let $\gcd(a, m) = \gcd(b, m) = 1$. This means that there exists x_0, y_0, x_1, y_1 such that $ax_0 + my_0 = 1,$ $bx_1 + my_1 = 1,$	
	which implies that $(ax_0 + my_0)(bx_1 + my_1) = 1$ $abx_0x_1 + m(ax_0y_1 + bx_1y_0 + my_0y_1) = 1.$ As such $\gcd(ab, m) = 1$.	
(ii)	We see that $a \equiv b \pmod{m}$ implies that for any c, $ac \equiv bc \pmod{m}.$ Also, since $c \equiv d \pmod{m}$, then $bc \equiv bd \pmod{m}$. Combining the two congruences give $ac \equiv bd \pmod{m}$.	
(iii)	Let $ax \equiv ay \pmod{m}$, this means that $m \mid a(x - y)$. Since $\gcd(a, m) = 1$, $m \nmid a$.	
	Consequently, by Euclid's lemma, $m \mid x - y, \text{ i.e. } x \equiv y \pmod{m}.$	
(iv)	Let $S = \{r_1, r_2, \dots, r_n\}$ be the reduced residue system modulo m. Since $\gcd(a, m) = 1$, for $1 \leq i \leq n$, $\gcd(ar_i, m) = 1 \text{ by (i)}$	
	Note that for $1 \leq i \leq n$, $\gcd(ar_i, m) = 1$ also implies that $aS = \{ar_1, \dots, ar_n\}$ has the same number of elements as S.	
	Finally, let $1 \leq i, j \leq n$, by considering the contrapositive of the 2nd bullet, we see that if $ar_i \equiv ar_j \pmod{m}$, then as $\gcd(a, m) = 1$, $r_i \equiv r_j \pmod{m} \text{ by (iii).}$ And hence, since r_i and r_j is in the same reduced residue system, $i = j$. Need to prove: If x is an integer such that $\gcd(x, m) = 1$, then $x \equiv ar_i \pmod{m} \text{ for some } ar_i \in aS ?$	

Commented [PWJ1]: I don't see how this is follows immediately. Perhaps a short proof would be required?

	Thus, aS must also be a reduced residue system modulo m .	
(v)	We see that 1,3,5,7,9,11,13,15 are numbers coprime to 16. As such $\phi(16)=8$.	
(vi)	Let $\{r_1, r_2, \dots, r_{\phi(m)}\}$ be a reduced residue system modulo m . By (iv), so is $\{ar_1, \dots, ar_{\phi(m)}\}$.	
	As both sets are of the same size, for each r_i there is one and only one ar_j such that $r_i \equiv ar_j \pmod{m}$.	
	This means that, by (ii), $\prod_{i=1}^{\phi(m)} r_i \equiv \prod_{j=1}^{\phi(m)} ar_j \pmod{m}$ $\prod_{i=1}^{\phi(m)} r_i \equiv a^{\phi(m)} \prod_{j=1}^{\phi(m)} r_i \pmod{m}.$	
	As $\gcd(r_i, m)=1$ for each i , it follows from (i) that $\gcd\left(\prod_{i=1}^{\phi(m)} r_i, m\right)=1$ as well.	
	Consequently, we again invoke the cancellation law, i.e. result (iii) to get $a^{\phi(m)} \equiv 1 \pmod{m}.$	
(vii)	By (v) and (vi), we have that	
	$3^8 \equiv 1 \pmod{16}$ $3^{8 \times 12 + 5} \equiv 3^5 \pmod{16}$ $3^{101} \equiv 243 \pmod{16}$ $\equiv 3 \pmod{16}.$ We see, therefore, that the remainder is 3.	